

how much Python you know. Python, incidentally, has a tremendous collection of libraries and third party APIs.

The advantage of starting off with Scikit-learn is that most programmers with similar aspirations have also started off along the same path. There is huge support for Scikit-learn with the entire code written using the Scikit-learn library, making the learning process fairly simple. It also helps that the official documentation of Scikit-learn is one of the more refined and well written. So, most of your doubts will probably get clarified by just going through the documentation.

To start off with Scikit Learn, just follow the instructions available at the official site, at <https://scikit-learn.org/stable/install.html>.

OpenCV

OpenCV is a cross platform library that lets the user tackle any kind of real-time computer vision.

Note: Computer vision is a field of computer science that deals with how computers can be made to acquire a high level knowledge from photos or videos. It includes the tasks of acquiring, processing, analysing and understanding the data.

OpenCV has a wide range of applications, starting with simple algorithms that deal with 2D images, all the way up to augmented reality, motion tracking, etc. OpenCV is not as user-friendly as Scikit-learn, yet, with a little time anyone can easily understand the flow of things – well enough to use OpenCV to its maximum extent. OpenCV, unlike Scikit-learn, is mostly written in C/C++ and is a cross platform library. This means that the users can truly optimise their code by using lower level and compiled languages such as C/C++ or Java. OpenCV also includes a statistical ML library with which it can stand alone to some extent. OpenCV also has integrations with leading DL frameworks

such as TensorFlow and Torch/PyTorch, making it a very versatile tool which can be used even if the provided ML library does not meet the user's demands.

One of the notable things about OpenCV is that it supports hardware acceleration in three different cases, namely:

- Intel's Integrated Performance Primitives
- A CUDA based GPU interface
- An OpenCL based GPU interface

The fact that OpenCV can use hardware acceleration means that the user can easily increase the performance of the code with simple modifications, thereby resulting in smoother real-time applications due to faster code execution.

Note: It is advisable to use hardware accelerations that are mathematically intensive to perform, else there is a chance of a drop in performance due to the time it takes to shift resources to the GPU from the CPU.

Detailed installation instructions for OpenCV are available from the official documentation for the Python programming language at https://docs.opencv.org/3.0-beta/doc/py_tutorials/py_setup/py_setup_in_windows/py_setup_in_windows.html. Other installations can also be found in the documentation.

TensorFlow

TensorFlow is an open source library which tackles data flow programming across a range of applications, from a symbolical mathematical library to the core library in neural network programming. Developed by the Google Brain team for the search giant's internal use, today it is used within the firm for purposes ranging from research to production.

Note: The TensorFlow library is so named because it operates on multi-dimensional data arrays called tensors, whilst performing calculations for neural networks.

Unlike the earlier libraries, TensorFlow has been streamlined to make neural networks and is optimised for DL applications. TensorFlow is best at data crunching at a scale where data can be realistically processed using any other older technique. You can do all this while keeping the usage so simple, that, any devoted Python user will feel right at home with the line-by-line implementation of the neural network, allowing the user to easily implement high level concepts that involve a huge amount of mathematical theory behind them, with the call of a single function. This allows the user to quickly create a neural model and focus on refining it rather than worry about the mathematics behind it, making for faster prototyping.

Like OpenCV, TensorFlow is not restricted to Python but supports other languages. Support ranges from having direct support to having third party APIs taking care of that. TensorFlow has a very flexible architecture system that allows it to be used in various devices, whether it is CPUs, GPUs or even TPUs (tensor processing units), as well as clustered servers or even handheld devices such as mobile phones. This allows TensorFlow to have a hardware acceleration capability that is unrivalled by most other libraries.

Note: Keras is a pure Python library unlike TensorFlow

TensorFlow can take code to the browser with the use of TensorFlow.js, making DL projects lightweight and easily scaleable. TensorFlow Lite, which was built for using TensorFlow, specifically for Android development (beginning from Android Oreo) is even lighter. It has a wide reach. There are several applications of this library for different use cases, from being the foundation of some new technologies such as Deep Dream, an automated image captioning software, to being